AI Roles in the Integrated Digital Tax System

Rwanda Revenue Authority · Real-Time Transaction Intelligence

Because the AI has full access to every transaction across all sellers simultaneously, it can monitor all merchants every second — something no human inspector could ever achieve. Below are the six core roles the AI performs within the system.

SYSTEM ARCHITECTURE DIAGRAM

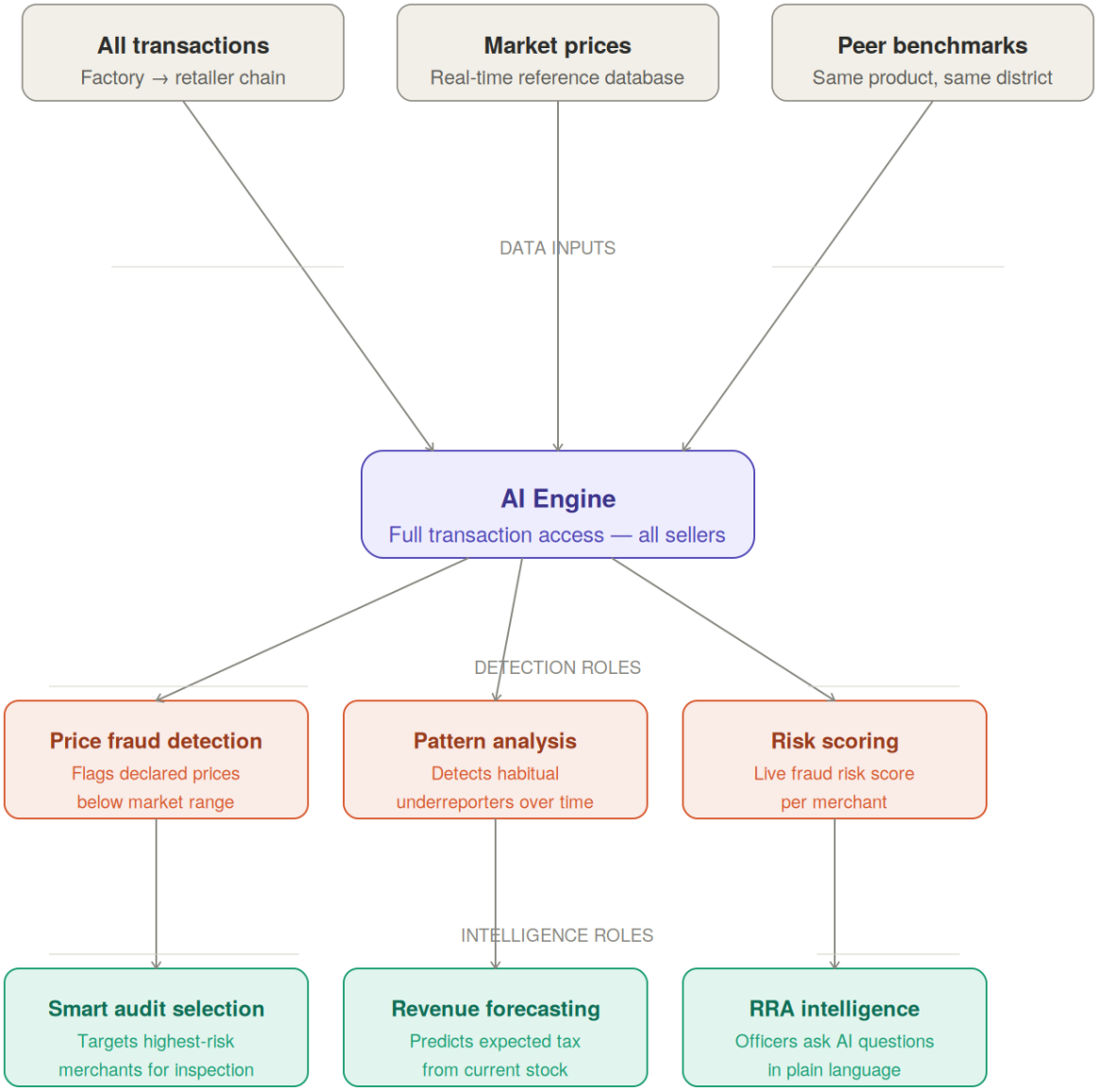


Figure 1 — AI Engine: data inputs, detection roles, and intelligence roles across the supply chain

DATA INPUTS

All Transactions
Full factory → wholesaler → retailer chain recorded in real time.

Market Prices
Live reference database of expected prices per product and district.

Peer Benchmarks
Declared prices of other merchants selling the same product nearby.

DETECTION ROLES

01 Price Fraud Detection

The moment a merchant enters a selling price, the AI immediately compares it against the known market price for that product, the purchase price already recorded in the system (declared price cannot legally be lower), and what nearby merchants are charging for the same product. If the declared price is suspiciously low, the transaction is flagged before it is approved — not after.

02 Behavioral Pattern Analysis

A single suspicious price could be a genuine mistake. But if a merchant consistently declares sugar at 1,500 RWF while every competitor in the same district declares 1,900–2,100 RWF, over weeks and months — that is a deliberate strategy, not an error. The AI tracks this pattern over time and builds a behavioral profile for every merchant. One-time anomalies are noted; repeated patterns trigger escalation.

03 Risk Scoring

Every merchant in the system receives a live fraud risk score, updated after every transaction. The score combines price deviation, tax payment history, customer complaint reports, and stock reconciliation accuracy. RRA can sort the entire merchant database by risk score at any moment and know exactly where to focus attention.

INTELLIGENCE ROLES

04 Smart Audit Selection

Instead of random inspections that waste resources and often miss real offenders, the AI tells RRA exactly which merchants to inspect and when. Physical audits become surgical rather than random — every inspector hour is directed at the highest-risk cases identified by transaction data, not by chance.

05 Revenue Forecasting

Because the AI sees all stock currently in the system — at every factory, wholesaler, and retailer — it can calculate the expected tax that should be collected when that stock is sold. If actual collected tax consistently falls below the forecast, the gap itself is evidence of underreporting, and the AI identifies which segment of the supply chain is causing it.

06 RRA Intelligence Interface

RRA officers can query the AI in plain language — for example: 'Which wholesalers in Kicukiro sold more than 500 KG of sugar last month but paid less than 50% of expected tax?' — and receive an instant, filtered answer backed by real transaction data. This transforms the entire transaction database into a live intelligence tool, not just a passive record archive.

The power of the AI comes from one fundamental fact: it never sleeps, never forgets, and sees every merchant simultaneously. A human inspector can audit one business per day. The AI monitors all of them, every second.